A6.5 Response before final destination

Any intermediate component that issues a transaction response before the transaction has reached its final destination, must ensure visibility of the transaction to any transactions from any upstream masters.

Note

In the context of this section, an upstream master is any device that accesses the intermediate component through a slave port on that component.

The requirements are:

- for accesses to all memory types, any subsequent transaction to the same or an overlapping address must observe the transaction for which the intermediate component issues a response
- for accesses to Device memory, the intermediate component must also maintain the ordering of any subsequent transaction with the same ID and to the same slave, relative to the transaction for which it issued a response.

An intermediate response can only be given to a transaction when the **AxCACHE** attribute indicates that it is permissible to do so.

The AXI protocol requires that the ordering guarantees for access to Device memory are a superset of the guarantees for accesses to Normal memory. This ensures that any transaction marked as Normal can be converted to Device without removing any of its original guarantees. To meet this requirement, the behavior for Device memory accesses to the same or overlapping addresses must be the same as for Normal memory accesses, regardless of the ID values.

Table A6-1 shows when ordering is required for all combinations of memory types, transaction IDs, and accesses to the same or overlapping addresses.

Table A6-1 Summary of ordering requirements

Memory type	Same ID	Overlapping address	Ordering required
Device	Yes	Yes	Yes
		No	Yes
	No	Yes	Yes
		No	No
Normal	Yes	Yes	Yes
		No	No
	No	Yes	Yes
		No	No

In Table A6-1, the entries for Normal memory apply, also, to transactions to cacheable memory. That is, they apply to all valid transactions for which **AxCACHE[3:1]** is not `0b000`.

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Chapter A7
Atomic Accesses

This chapter describes the AXI4 concept of single-copy atomicity size and how the AXI protocol implements exclusive access and locked access mechanisms. It contains the following sections:

- *Single-copy atomicity size* on page A7-90
- *Exclusive accesses* on page A7-92
- *Locked accesses* on page A7-95
- *Atomic access signaling* on page A7-96.

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